Introduction…

- The purpose of distribution system is to deliver water to consumer with appropriate quality, quantity and pressure.

- Distribution system is used to describe collectively the facilities used to supply water from its source to the point of usage.

Requirements of Good Distribution System...

- Water quality should not get deteriorated in the distribution pipes.
- It should be capable of supplying water at all the intended places with sufficient pressure head.
- It should be capable of supplying the requisite amount of water during fire fighting.

- The layout should be such that no consumer would be without water supply, during the repair of any section of the system.

- All the distribution pipes should be preferably laid one metre away or above the sewer lines.

- It should be fairly water-tight as to keep losses due to leakage to the minimum.

Layouts of Distribution Network

- The distribution pipes are generally laid below the road pavements, and as such their layouts generally follow the layouts of roads.

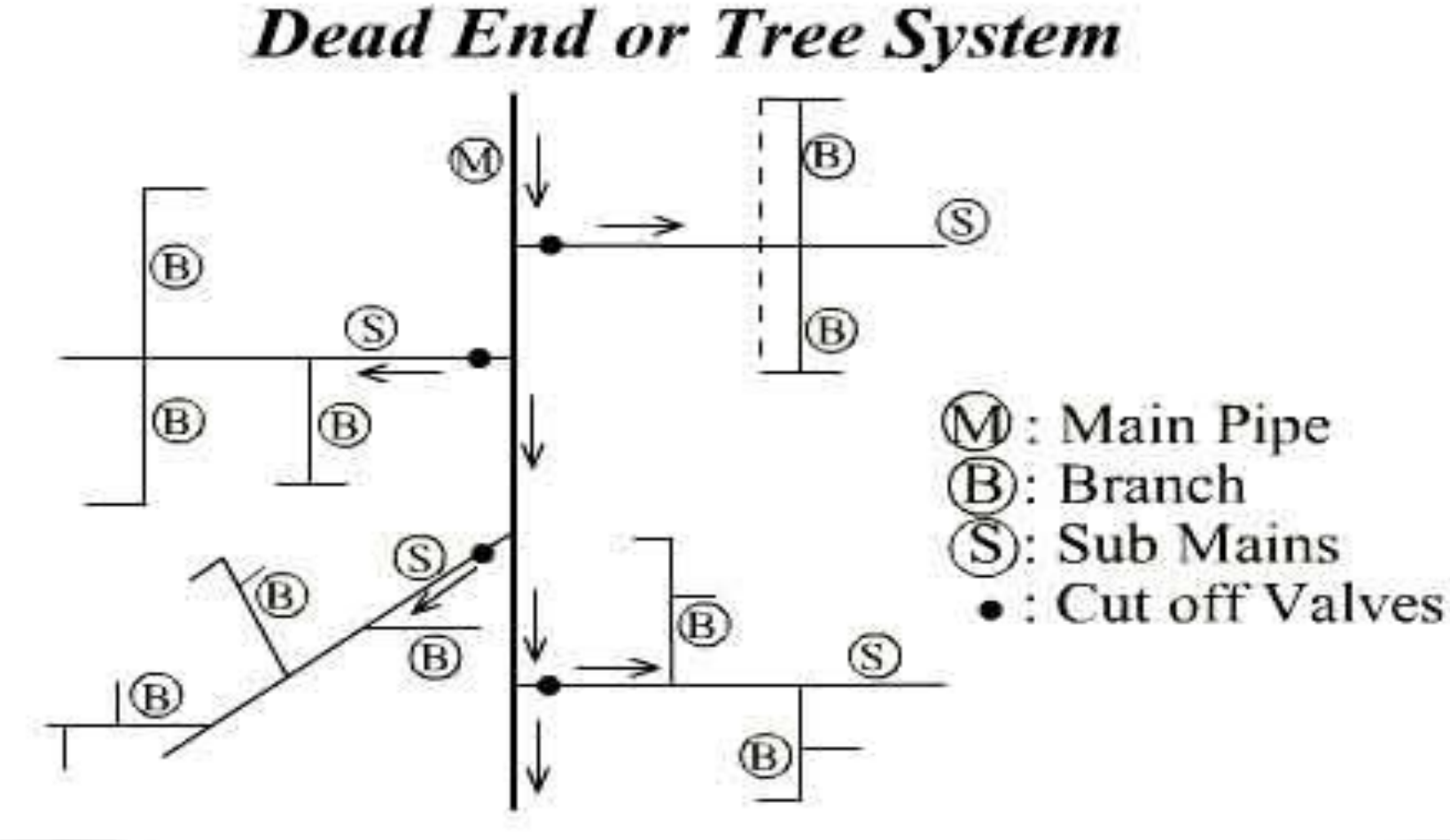
- There are, in general, four different types of pipe networks; any one of which either singly or in combinations, can be used for a particular place.

They are:

- ❖Dead End System
- ❖Radial System
- ❖Grid Iron System
- ❖Ring System

Dead End System...

- It is suitable for old towns and cities having no definite pattern of roads.



Advantages

- Relatively cheap.
- Determination of discharges and pressure easier due to less number of valves.

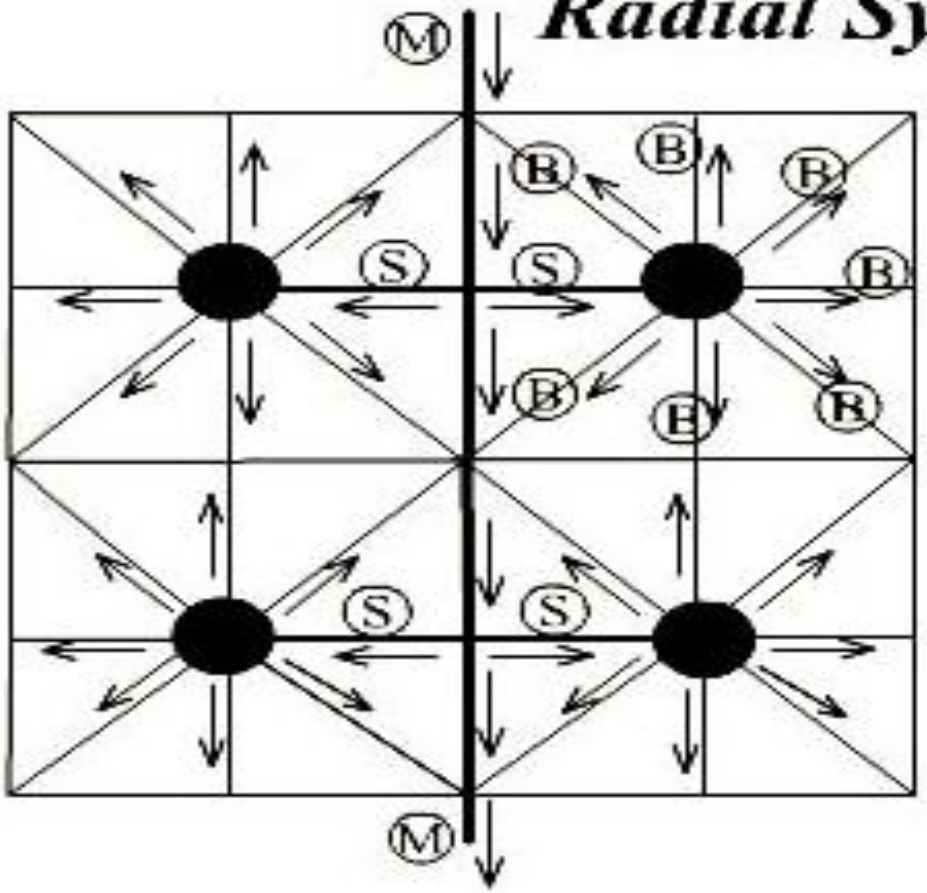
Disadvantages

- Due to many dead ends, stagnation of water occurs in pipes.

Radial System...

- The area is divided into different zones.
- The water is pumped into the distribution reservoir kept in the middle of each zone.
- The supply pipes are laid radially ending towards the periphery.

Radial System



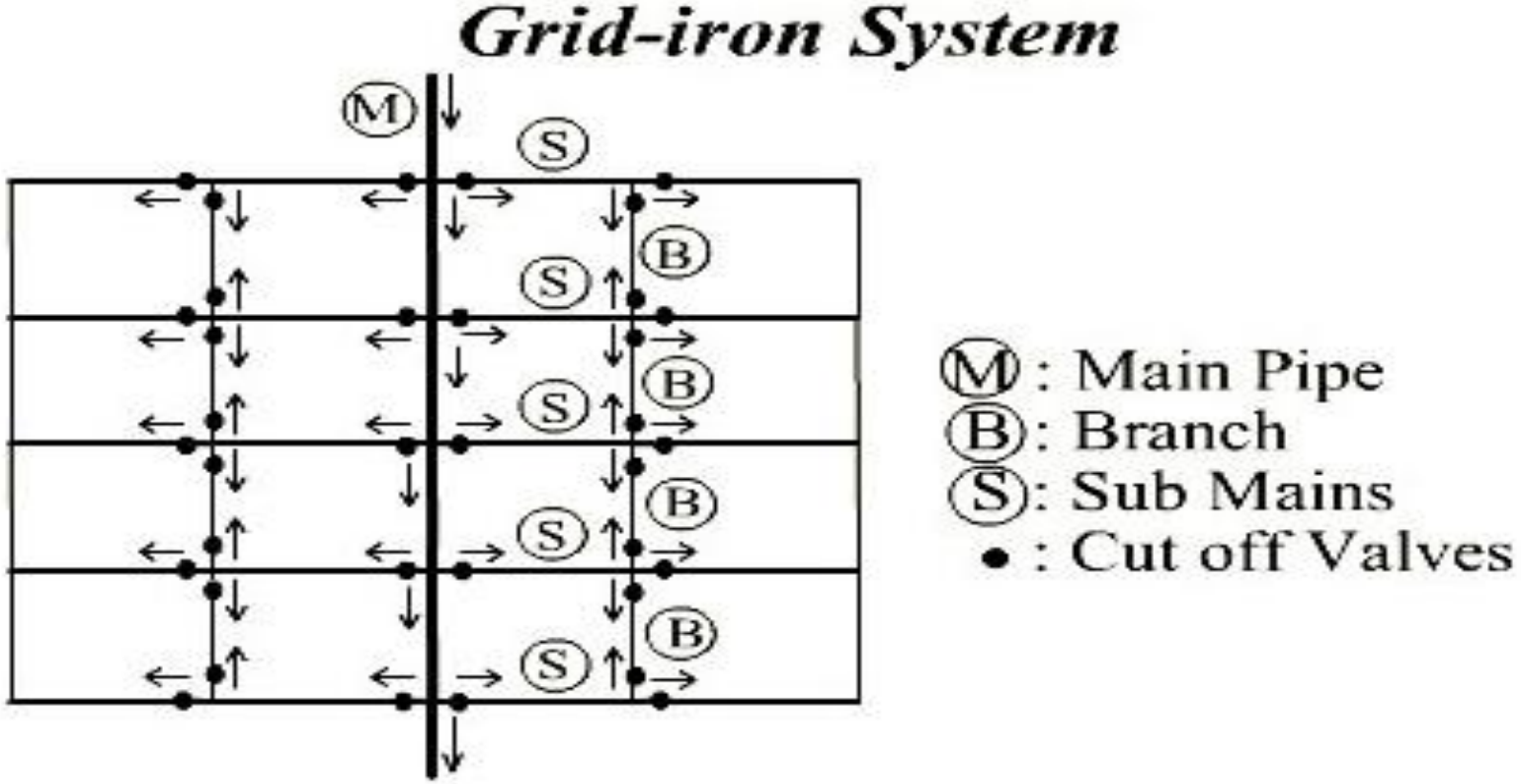
: Distribution Reservoirs

Advantages:

- It gives quick service.
- Calculation of pipe sizes is easy.

Grid Iron System...

- It is suitable for cities with rectangular layout, where the water mains and branches are laid in rectangles.



Advantages

- Water is kept in good circulation due to the absence of dead ends.
- In the cases of a breakdown in some section, water is available from some other direction.

Disadvantages

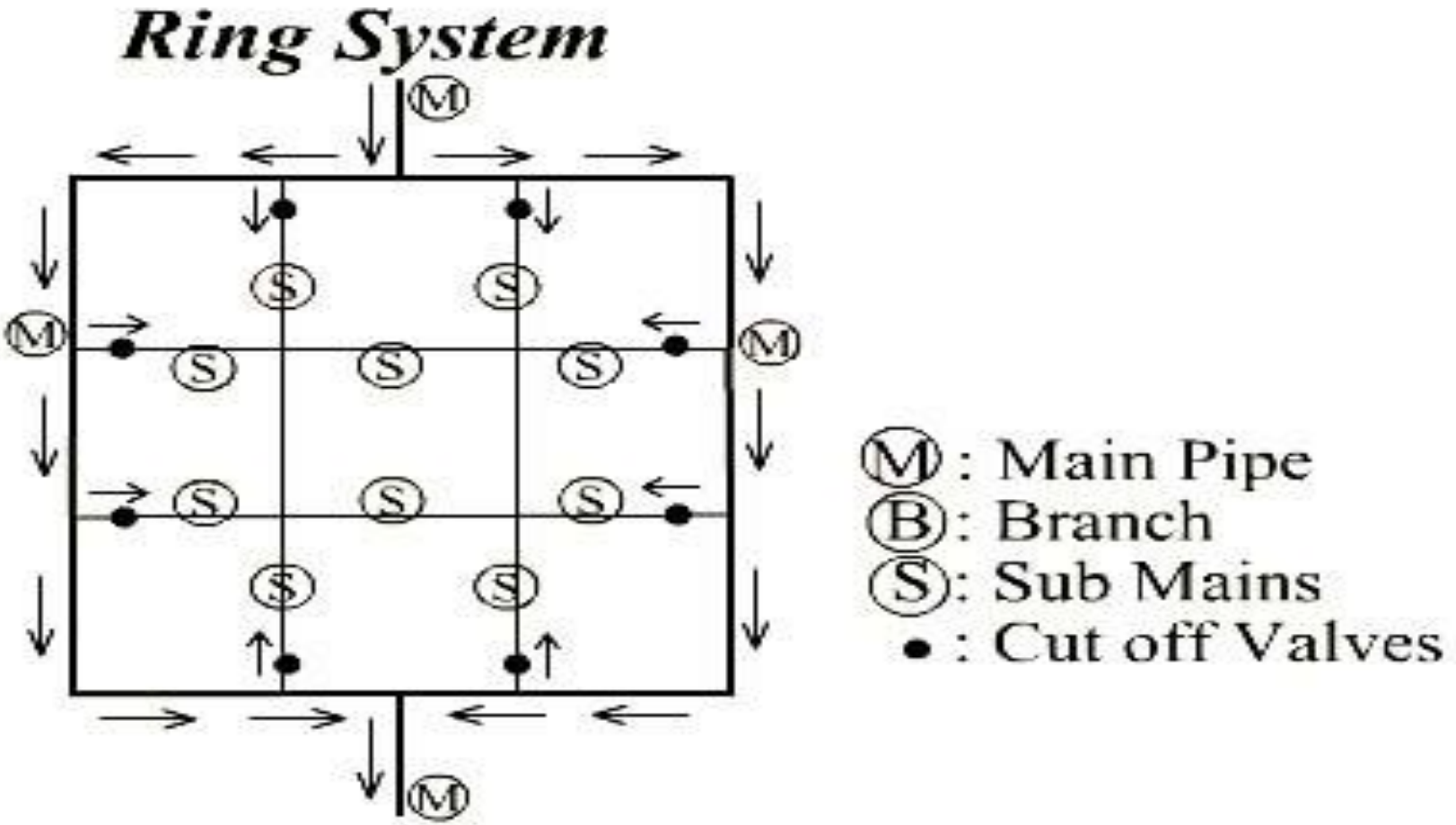
- Exact calculation of sizes of pipes is not possible due to provision of valves on all branches.

Ring System...

- The supply main is laid all along the peripheral roads and sub mains branch out from the mains.
- This system also follows the grid iron system with the flow pattern similar in character to that of dead end system.
- So, determination of the size of pipes is easy.

Advantages

- Water can be supplied to any point from at least two directions.

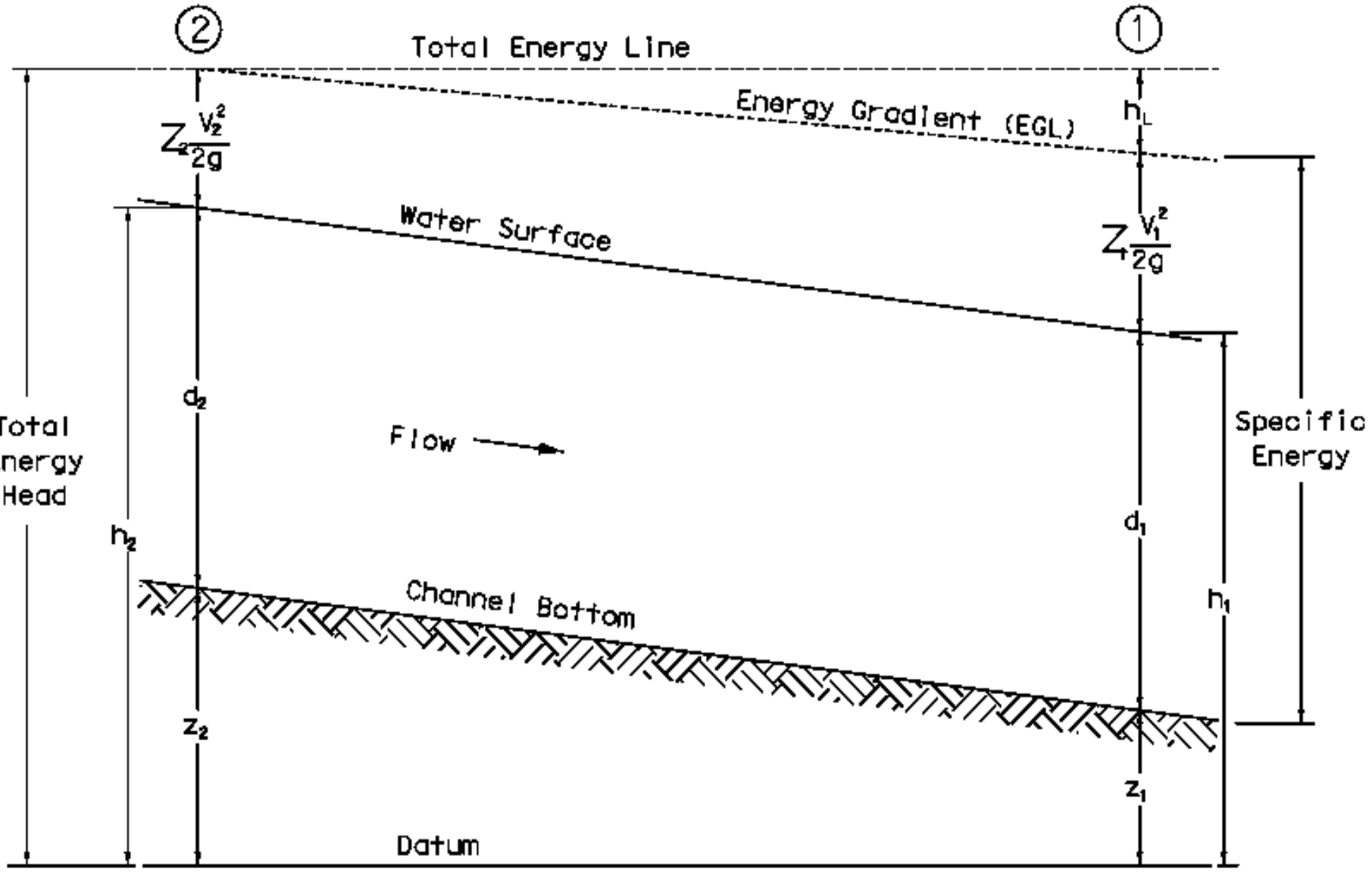


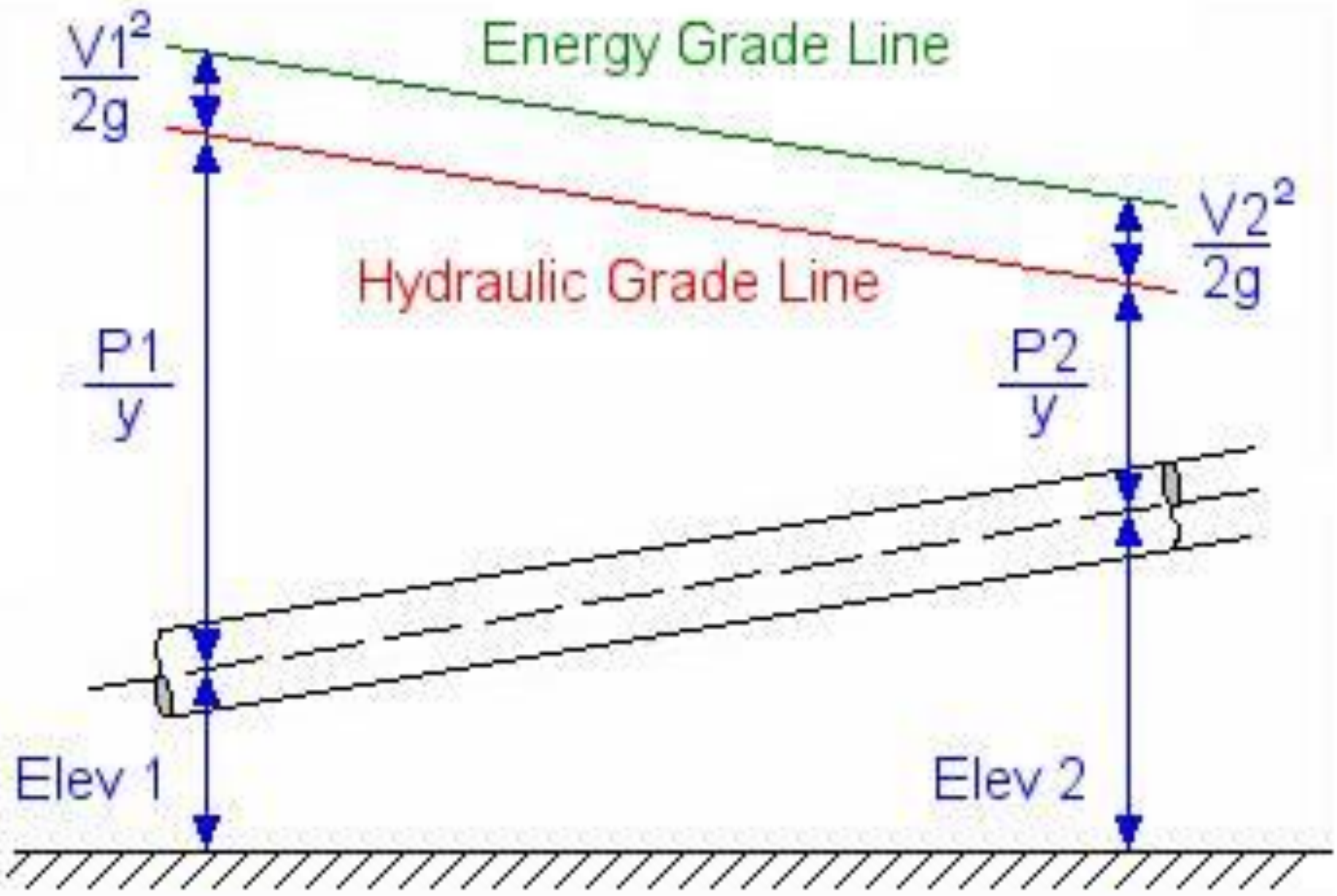
Methods of water distribution...

- For efficient distribution system adequate water pressure required at various points.

- Depending upon the level of source, topography of the area and other local conditions the water may be forced into distribution system by following ways -

1. Gravity system
2. Pumping system
3. Combined gravity and pumping system


②
①
Total Energy Line
Energy Gradient (EGL)
h_L
$Z_2 \frac{V_2^2}{2g}$
$Z_1 \frac{V_1^2}{2g}$
Water Surface
d_2
Flow
Total Energy Head
Specific Energy
h_2
d_1
h_1
Channel Bottom
z_2
z_1
Datum


$\frac{V1^2}{2g}$
Energy Grade Line
$\frac{V2^2}{2g}$
Hydraulic Grade Line
$\frac{P1}{y}$
$\frac{P2}{y}$
Elev 1
Elev 2

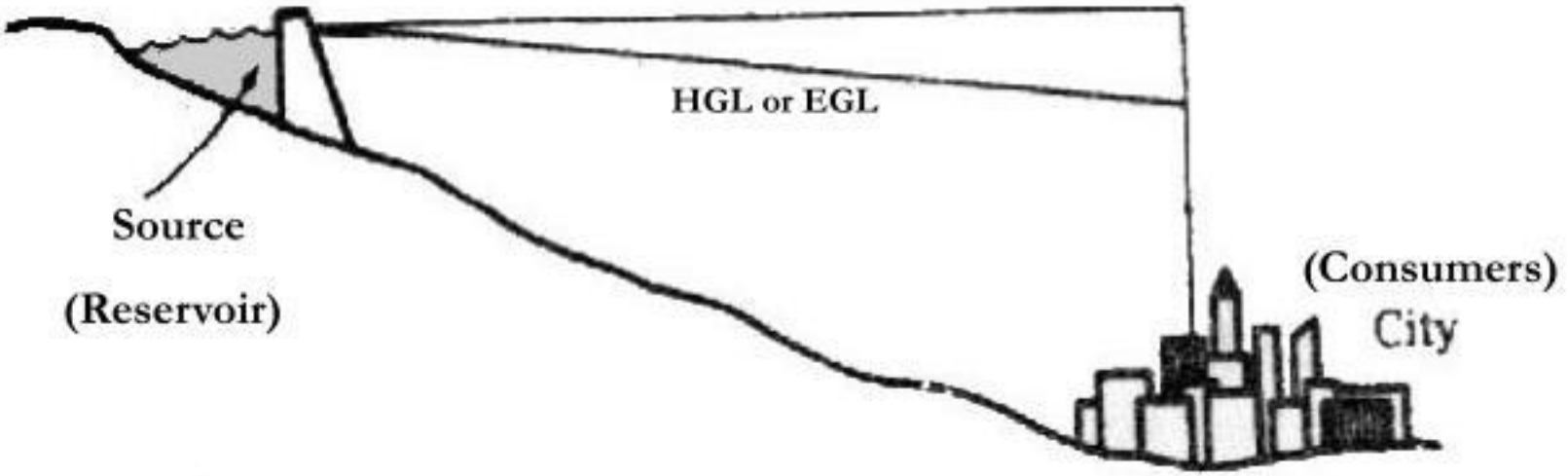
Gravity system…

- Suitable when source of supply is at sufficient height.
- Most reliable and economical distribution system.
- The water head available at the consumer is just minimum required.
- The remaining head is consumed in the frictional and other losses.

Gravity Supply

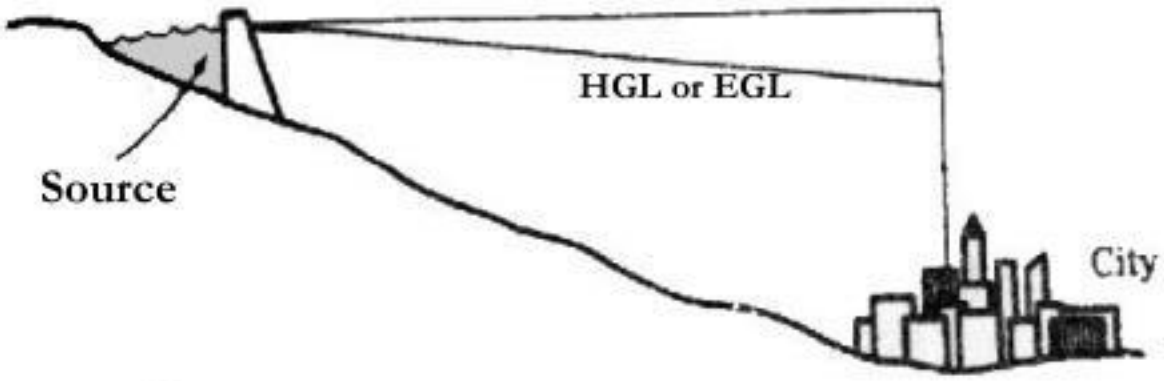
- The **source** of supply is at a sufficient elevation **above** the **distribution area** (consumers).

so that the desired pressure can be maintained

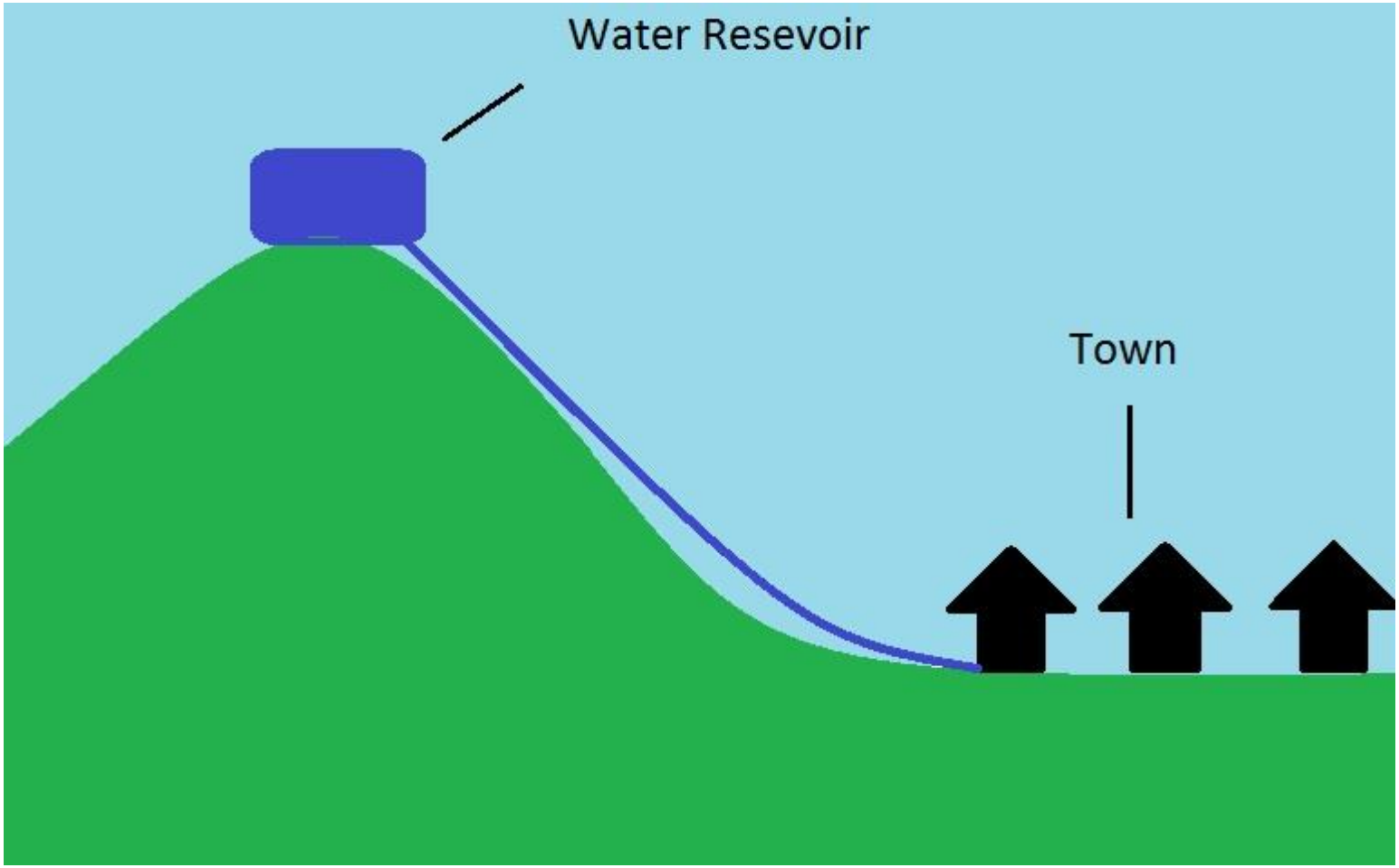


Gravity-Supply System

Advantages of Gravity supply



- No energy costs.
- Simple operation (fewer mechanical parts, independence of power supply, ….)
- Low maintenance costs.
- No sudden pressure changes


Water Resevoir
Town

Pumping system…

- Treated water is directly pumped in to the distribution main with out storing.
- Also called pumping without storage system.
- High lifts pumps are required.
- If power supply fails, complete stoppage of water supply.
- This method is not generally used.

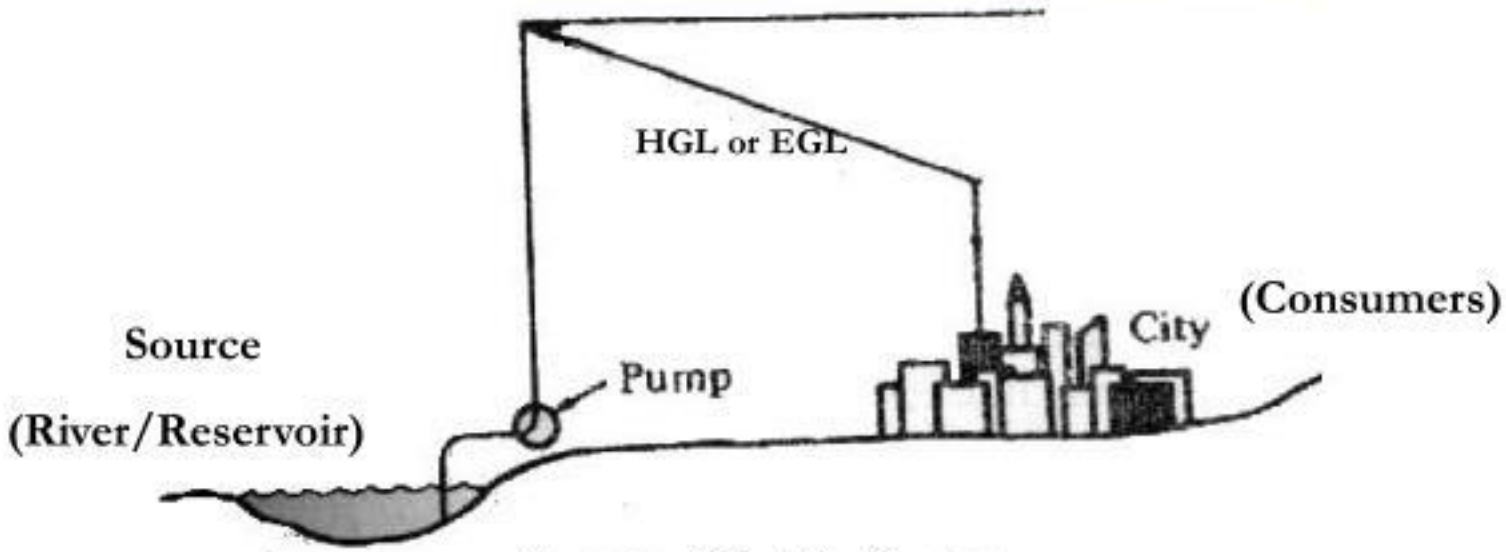
Water Distribution Systems

Pumped Supply

➔ Used whenever:

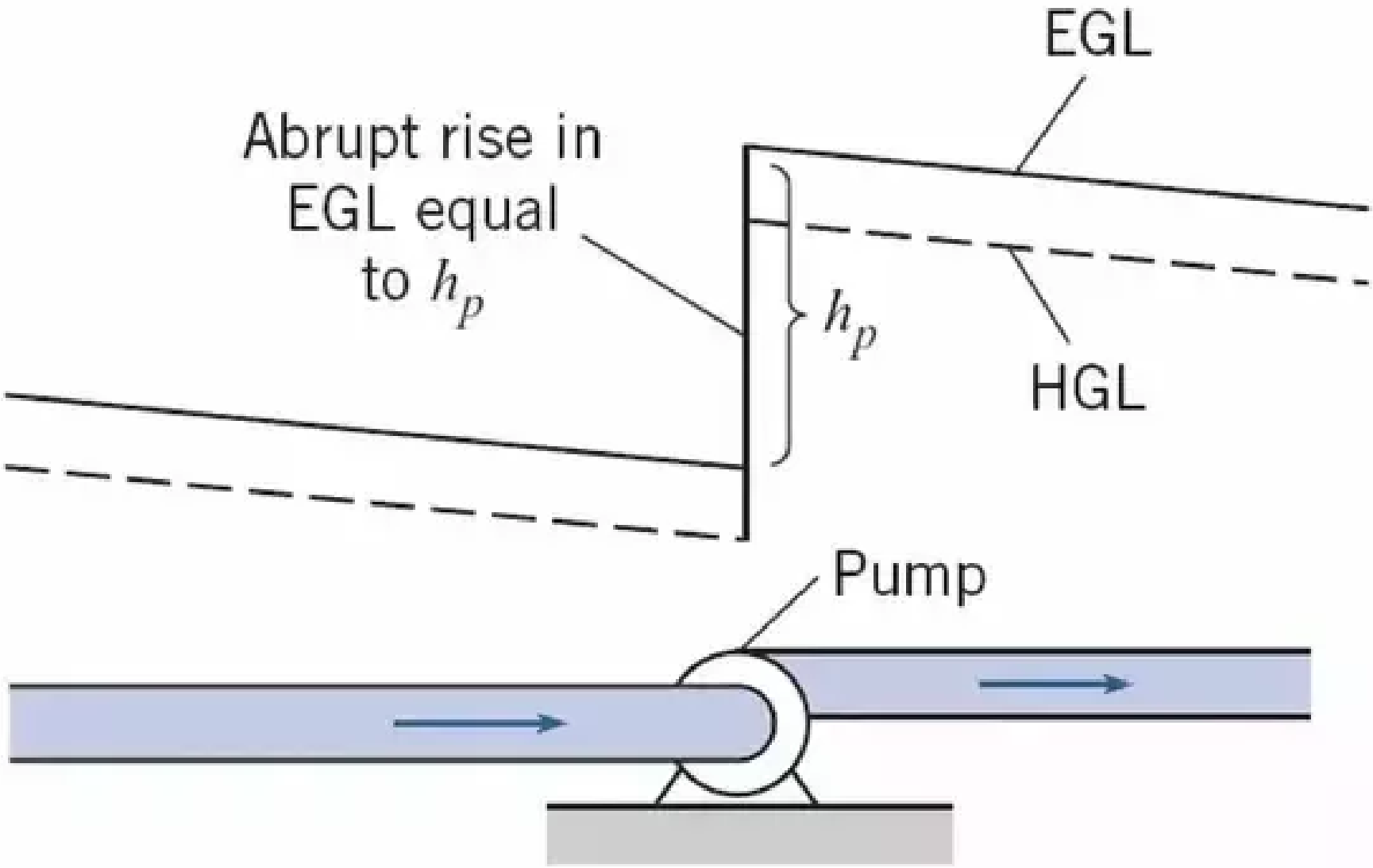
- The source of water is lower than the area to which we need to distribute water to (consumers)
- The source cannot maintain minimum pressure required.

➔ pumps are used to develop the necessary head (pressure) to distribute water to the consumer and storage reservoirs.



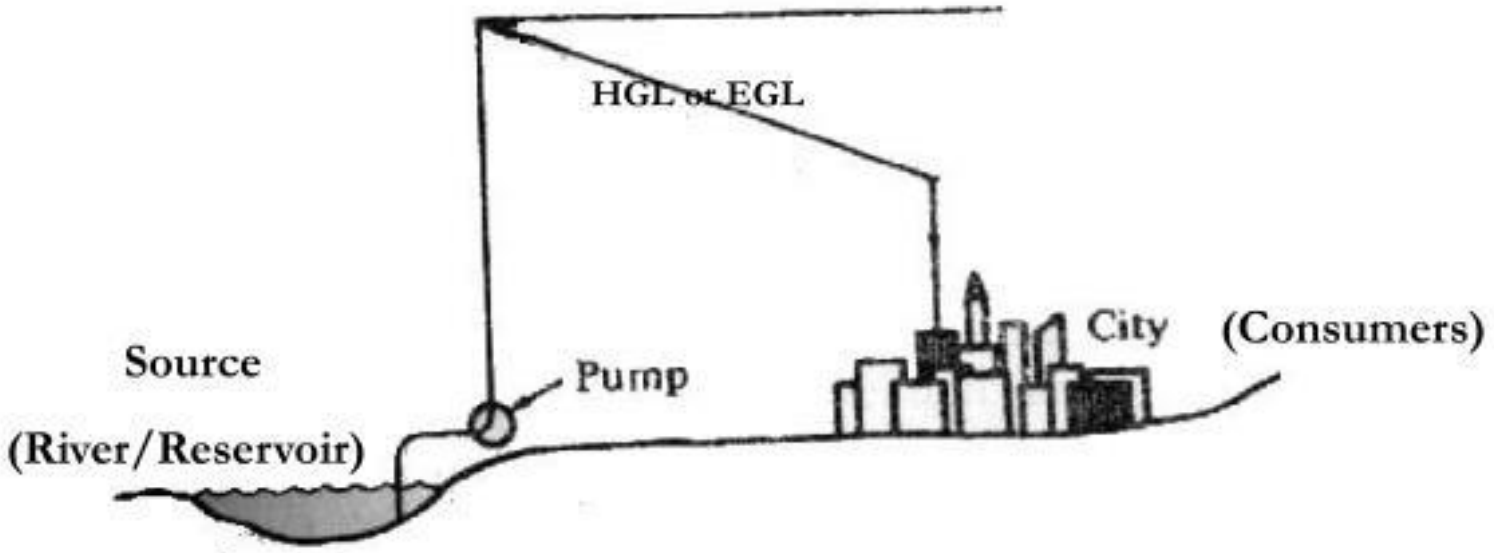
Pumped-Supply System

Grade Line Interpretation - Pump



Disadvantages of pumped supply

- ☒ Complicated operation and maintenance.
- ☒ Dependent on reliable power supply.
- ☒ Precautions have to be taken in order to enable permanent supply:
  - Stock with spare parts
  - Alternative source of power supply ….



Combined gravity and pumping system

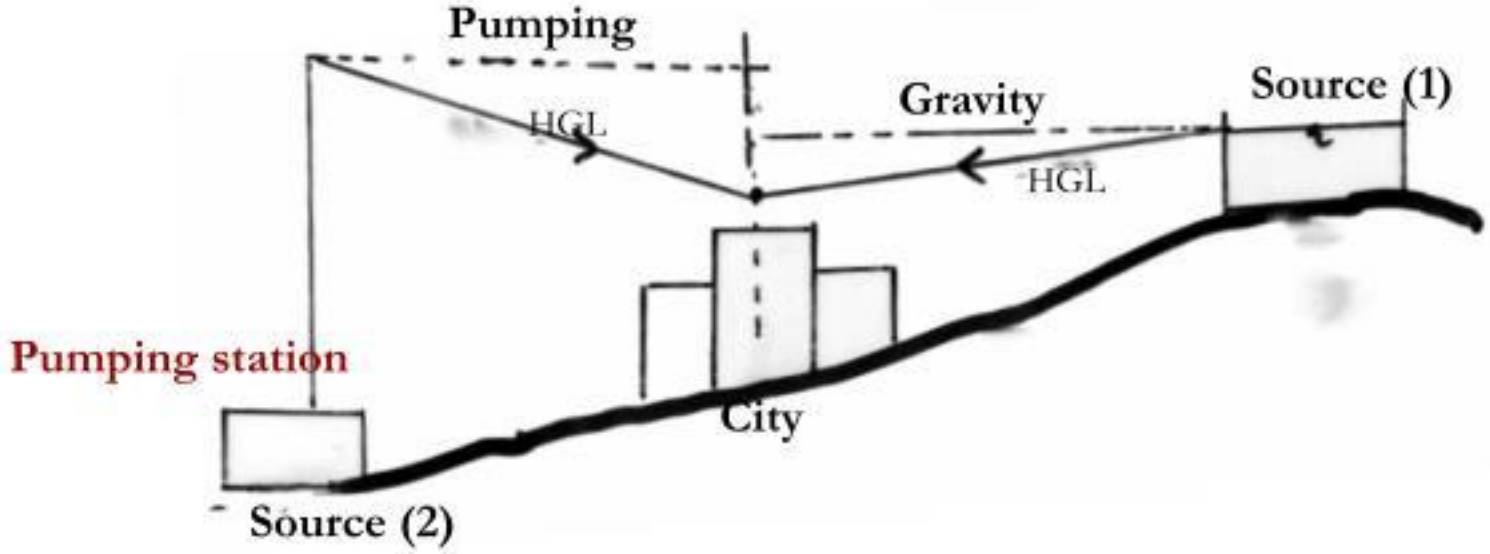
- Most common system.
- Treated water is pumped and stored in an elevated distribution reservoir.
- Then supplies to consumer by action of gravity.
- The excess water during low demand periods get stored in reservoir and get supplied during high demand period.
- Economical, efficient and reliable system.

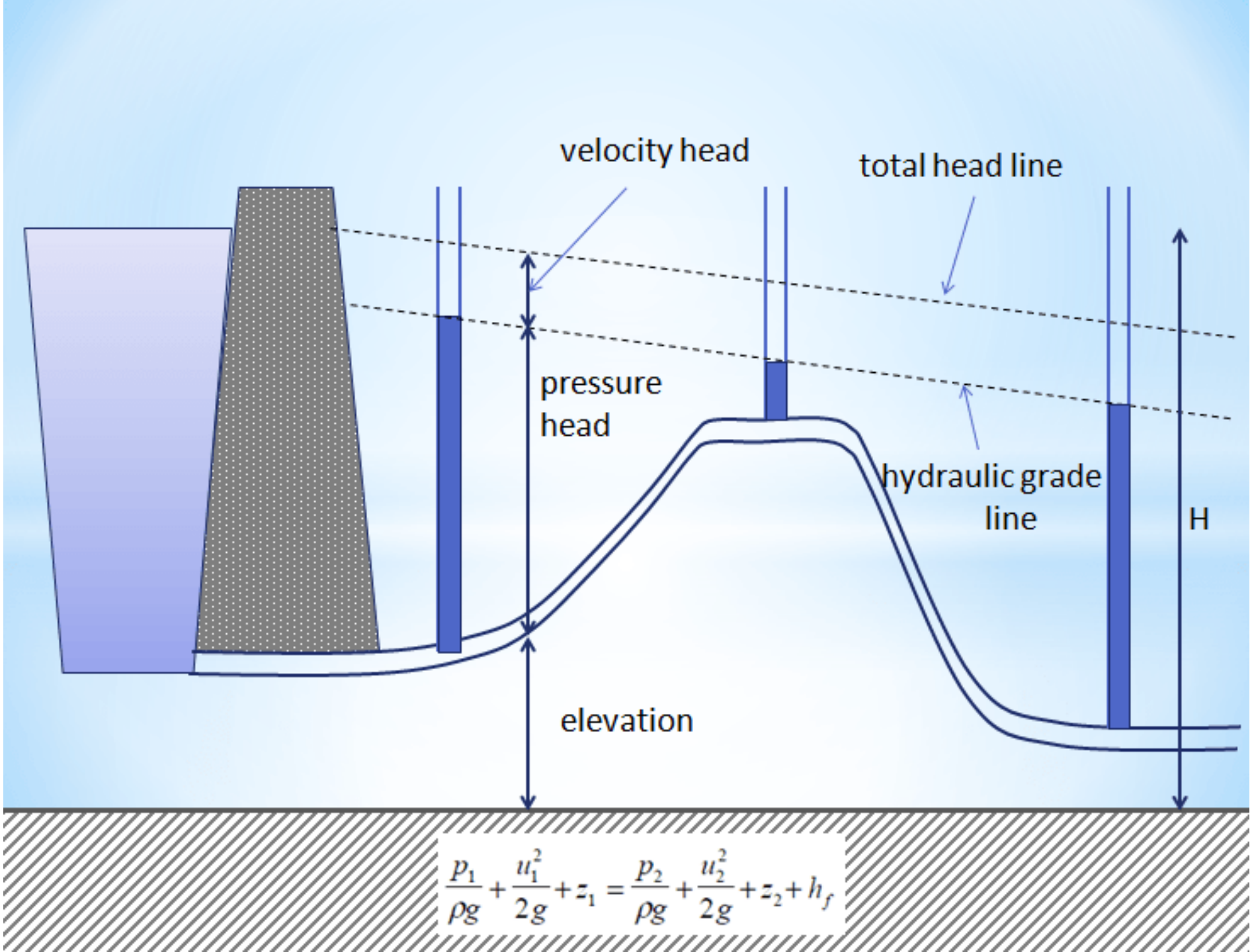
Combined Supply

(pumped-storage supply)

- Both **pumps** and **storage reservoirs** are used.
- This system is usually used in the following cases:

1) When two sources of water are used to supply water:





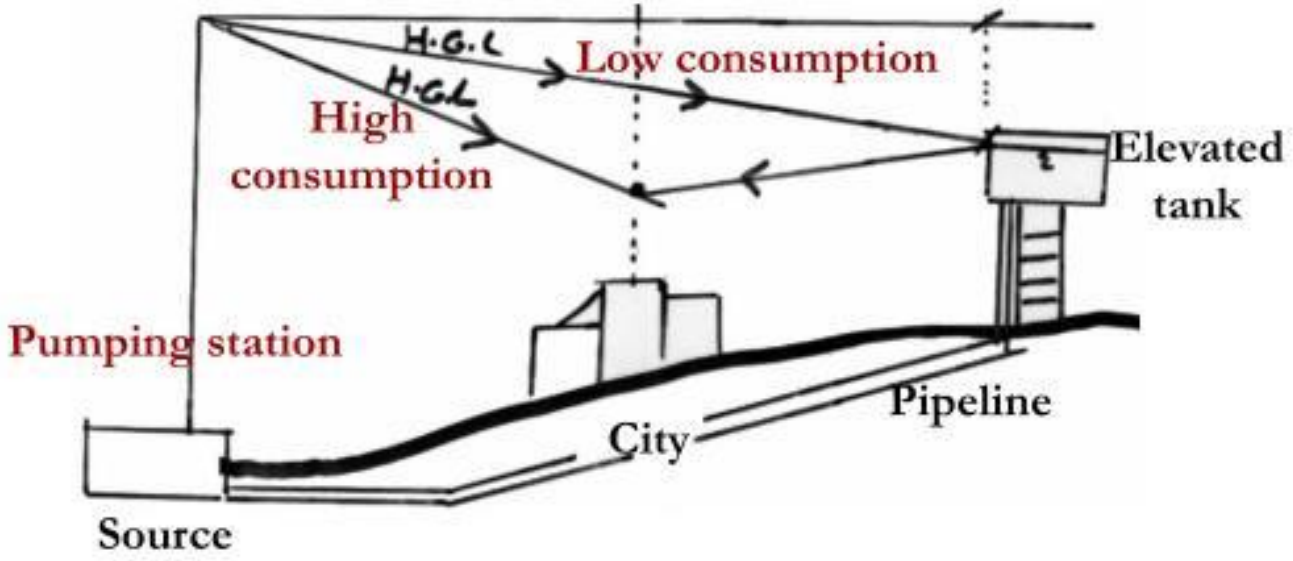
$$\frac{p_1}{\rho g}+\frac{u_1^2}{2g}+z_1=\frac{p_2}{\rho g}+\frac{u_2^2}{2g}+z_2+h_f$$

Water Distribution Systems

Combined Supply (Continue)

2) **In the pumped system sometimes a storage (elevated) tank is connected to the system.**

- When the water consumption is low, the residual water is pumped to the tank.
- When the consumption is high the water flows back to the consumer area by gravity.



Water Distribution Systems

Combined Supply (Continue)

3) When the source is lower than the consumer area

- A tank is constructed above the highest point in the area,
- Then the water is pumped from the source to the storage tank (reservoir).
- And the hence the water is distributed from the reservoir by gravity.

